ChatGPT

Momentum-Based Weekly Equity Strategy

We design a long-only cross-sectional momentum strategy over 200 stocks (plus a benchmark index) with weekly rebalances. At each rebalance we compute a **momentum score** for each stock using its recent price action and volatility. We then rank stocks by this score and hold at most 6 names. Crucially, the strategy **scales exposure to volatility** and applies filters (trend confirmation, 52-week high, etc.) to avoid churn. Academic research shows pure momentum can suffer large "crashes" after market sell-offs and volatility spikes [1]. To mitigate this, we adopt volatility-targeting and additional filters: for example, targeting a fixed portfolio volatility (e.g. 10–12% annual) has been shown to nearly eliminate momentum crashes and roughly **double Sharpe ratio** [2]. In practice, this means reducing position sizes in high-volatility periods and requiring stronger trend signals before buying. The key elements are:

- **Price-momentum signals:** Compute trailing returns (e.g. 21-day, 63-day, 252-day returns). Research confirms a stock's own past returns (e.g. 12-month return) are predictive of its future performance [3]. We combine multiple horizons (short and medium term) to capture both recent and intermediate trends.
- **Risk-adjusted scoring:** We normalize returns by recent volatility (or directly allocate "equal-risk" weights). For example, one can divide a stock's return by its standard deviation or set position sizes ∝ 1/volatility [4] [5]. This ensures volatile winners don't dominate and keeps each position's risk in check. Empirically, volatility-scaled momentum strategies have shown far lower drawdowns [2].
- **Trend filters:** To avoid buying in a range, we only score stocks in uptrends. For instance, require the current price to exceed its 50-day moving average or have RSI(14) above 50. This confirms market participation by buyers. If these conditions fail, we give the stock a very low (or negative) score.
- **52-week high filter:** We compute `near_high = Price / 52-week_high`. Recent research finds stocks very close to their highs tend to form momentum crash patterns [6]. We can **penalize stocks with high** `near_high` (e.g. multiply score by 1 – near_high) or skip those above a threshold. This "near-high" adjustment helps the portfolio avoid crowded breakouts that often reverse in choppy markets.
- **Volume/liquidity check:** (Optional) Exclude ultra-low-volume stocks to ensure tradability. A simple rule: require today's volume ≥ 3-month average volume. This filter avoids signals on illiquid stocks, which can whipsaw in ranges.

Ranking and Selection

1. **Compute scores:** For each stock, calculate a composite momentum score. For example:
2. Let `mom_short` = (Price_now / Price_21d_ago – 1), `mom_long` = (Price_now / Price_63d_ago – 1).
3. Let `vol` = standard deviation of daily returns over the past 63 days.
4. Define an un-normalized score `s = w1*mom_short + w2*mom_long` (weights w1,w2 set by backtest or equal).
5. Risk-adjust by volatility: e.g. final_score = s / vol.

6. Apply filters: if Price<MA(50) or RSI<50, set final_score=–∞. Compute `near_high = Price / (max Price over 252d)` and adjust score = final_score * (1 – near_high).

7. **Rank stocks by score:** Sort all eligible stocks by descending score. (If desired, one could also require score > 0.)

8. **Select top stocks:** Take the highest scores, up to 6 names. If fewer than 6 stocks have positive signals, invest in the ones available (keeping the rest in cash). We may also impose sector or industry caps to ensure diversification, but the core rule is "top 6 momentum leaders."

9. **Position weights:** Weight the selected stocks to equalize risk. A simple approach is **inverse-volatility weighting**: allocate weight ∝ (1/σ_stock), then normalize so weights sum to 100%. This gives each stock the same volatility contribution [5] [4]. For example, if two stocks have volatilities 20% and 10% (annualized), they might receive weights 5% and 10% respectively (because 1/20 = 0.05, 1/10 = 0.10, which after normalization equalize risk). This ensures that high-vol stocks get smaller size.

10. **Overall risk target:** Optionally, scale the entire portfolio so that total volatility is near a fixed level (e.g. 10% annual). This is akin to Barroso & Santa-Clara's "constant volatility 12%" scheme [2]. In practice, we can estimate portfolio vol each week (based on stock vols and correlations) and scale all weights up/down to target. Constant- or conditional- volatility schemes have been shown to **significantly cut drawdowns**. For instance, reducing exposure during high-vol states (and increasing in low-vol) "significantly reduced drawdowns and tail risk" in major equity markets [7].

Position Sizing & Risk Controls

- **Equal-risk weights:** As above, size each position by inverse volatility so each contributes equally to risk [5]. This guards against any one position blowing out in choppy markets. It also automatically shrinks sizes in volatile stocks and periods.
- **Position caps:** Impose a maximum weight per stock (e.g. ≤20%) and/or per sector. Since we only hold 6 stocks, equal-risk often implies weights ≲15–17%, but we double-check no single name dominates beyond the cap.
- **Stop-loss / exit rules:** We may implement simple stops or trailing stops to cut losses. For example, exit a position if it falls more than X% (e.g. 5–10%) from entry, or breaks below its 10-day low. Trailing stops protect against sharp reversals during choppy spikes. (While stops are not directly backed by citations here, they are common risk controls in momentum systems.)
- **Regime filter (optional):** If the benchmark index shows sustained weakness, we reduce exposure. For instance, require the index's own momentum (say 63-day return) to be positive before full investment; if not, only 50% capital is deployed (holding 50% in cash). Empirically, momentum strategies often struggle in down markets [1], so tying our aggressiveness to the overall trend can reduce drawdowns.

Overall, these risk controls draw directly on research: volatility-managed momentum (scaling by 12% vol) "virtually eliminated crashes" and doubled Sharpe [2], while conditional volatility targeting further cuts drawdown [7]. By sizing down positions when volatility is high, the strategy keeps losses moderate even if individual stocks whip around [5].

Adaptation to Choppy Markets

In sideways or range-bound conditions, few stocks exhibit strong trend, so the strategy naturally holds fewer or smaller positions. The combination of our filters and sizing achieves robustness:

- **Selective entries:** Requiring trend confirmation (e.g. price > MA) means we only trade when a stock shows real momentum, not just noise. In a horizontal market, fewer stocks clear these filters, so we trade less, reducing false signals.
- **Volatility scaling:** Choppy markets usually mean higher volatility. Our volatility-based sizing automatically **reduces position sizes** when markets are more erratic [5]. In effect, we dial down risk during turbulent sideways periods. As Barroso & Santa-Clara found, reducing exposure in high-volatility environments nearly eliminated momentum crashes [2].
- **52-week filter:** Many choppy-momentum crashes occur when a broad rally causes lagging stocks (far from highs) to snap back [6]. By penalizing or avoiding stocks too close to their 52-week highs, we lessen the chance of being on the wrong side of a rebound. This “near-high neutrality” was shown to significantly attenuate crash risk [6].
- **Concentration on leaders:** Limiting to the top 6 keeps the portfolio focused on the strongest signals. A small, high-conviction basket tends to outperform a broad, dilute one in range-bound markets, because we aren’t averaging in many marginal trades.
- **Market filter (if used):** By reducing overall exposure when the index momentum is negative, we avoid full-power trading in a non-trending market, further curbing drawdowns.

Together, these features mean the strategy takes a conservative posture during choppiness. When momentum is weak or conditions lack a clear trend, the approach will have low leverage and avoid marginal trades, resulting in **smaller drawdowns**. In contrast, when a genuine uptrend emerges, the filters are satisfied by more stocks and vol-sizes expand (since volatility often falls in trending markets), allowing the strategy to capture strong momentum gains. This dynamic (high exposure in low-vol trends, low exposure in high-vol noise) is exactly what the literature recommends [2] [7].

Example Pseudocode

```
# Weekly rebalance logic (illustrative pseudocode):

# 1. Compute indicators for each stock
for stock in universe:
    price = current_price(stock)
    mom_short = price / price_21d_ago(stock) - 1
    mom_long  = price / price_63d_ago(stock) - 1
    vol       = stddev(returns_last63d(stock)) * sqrt(252)  # annualized
    # Trend filter: require uptrend
    if price < moving_average(stock, 50):
        score = -1e9
    else:
        # Base momentum score (weighted combination of horizons)
        score = 0.6*mom_short + 0.4*mom_long
        # Normalize by volatility
```

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        score = score / vol
    # Near-high adjustment
    high52 = max(price over past 252 trading days)
    near_high = price / high52
    score = score * (1.0 - near_high)   # penalize stocks close to 52w high
    scores[stock] = score

# 2. Select stocks
sorted_stocks = sort_by_value_desc(scores)
selected = []
for stock in sorted_stocks:
    if scores[stock] <= 0:
        break
    selected.append(stock)
    if len(selected) == 6:
        break

# 3. Position sizing (inverse-vol)
weights = {}
inv_vols = [1.0 / stddev(returns_last63d(s)) for s in selected]
sum_inv = sum(inv_vols)
for i, stock in enumerate(selected):
    weights[stock] = inv_vols[i] / sum_inv

# 4. Risk check (optional)
# If benchmark momentum is negative, scale weights down
bench_mom = index_price / index_price_63d_ago - 1
if bench_mom < 0:
    for stock in weights:
        weights[stock] *= 0.5  # half the exposure

# 5. Apply stop-loss logic (in live trading, monitor and sell if triggered)
```

In summary, this strategy blends **straightforward momentum signals** with **volatility-based risk controls**. It ranks stocks by their risk-adjusted trailing returns, selects the top handful, and sizes them inversely to volatility [3] [5]. By incorporating trend filters and a 52-week high adjustment, it reduces whipsaw trades. These design choices are grounded in research: volatility scaling and conditional risk management have been shown to halve momentum crashes and boost Sharpe ratios [2] [7]. Overall, the strategy aims for strong returns in trending markets while naturally cutting exposure in choppy or sideways conditions, achieving lower drawdowns without sacrificing upside.

Sources: Academic and practitioner research on momentum and risk management, including studies on volatility-targeted momentum [2] [8] [7] [6] and time-series momentum [3] [4], along with risk-control principles [1] [5].

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